HAT-P-36b

R-1936

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-36_250612` · created 2026-08-06T14:05:10+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460838.7283 +/- 0.0048 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.13 +/- 0.034
Transit depth (Rp/Rs)^2	1.68 +/- 0.89 [%]
Orbital Inclination (inc)	87.05 +/- 2.66
Airmass coefficient 1 (a1)	0.946 +/- 0.014
Airmass coefficient 2 (a2)	0.033 +/- 0.015
Scatter in the residuals of the lightcurve fit is	1.52 %
Best Comparison Star	#2 - [168, 399]
Optimal Method	PSF photometry
Transit Duration (day)	0.0904 +/- 0.0052

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.13 ± 0.034 vs 0.126 (deviation 0.12σ)
Duration fit vs published	0.0904 d vs 0.0925 d (deviation 0.4σ)
Mid-time O–C	-3.7 min
Depth SNR	3.8
Residual scatter	1.52 %

Light curve

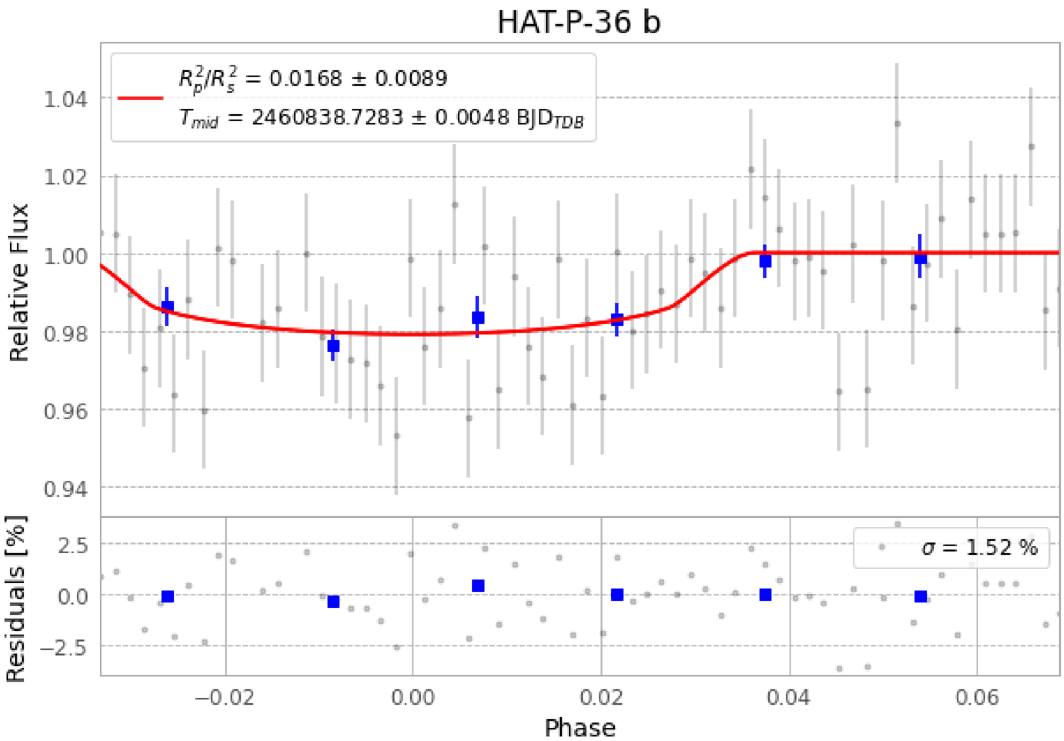

Figure 1 — FinalLightCurve_HAT-P-36 b_2025-06-11.png (SHA-256 82ee600b9dca2ba1…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [311, 252], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 c5bbd6a1ed94767e…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit d8a7143`

Data provenance

66 FITS frames (43 MB), `HATP-36250612042415.FITS` → `HATP-36250612073916.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-250612.log (e703bd87585e…), FinalLightCurve_HAT-P-36 b_2025-06-11.png (82ee600b9dca…), FinalLightCurve_HAT-P-36 b_2025-06-11.csv (df354cfd3ab6…)

Compute resources

Wall time 126.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)